

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

MECHANICAL PROPERTIES OF FLUIDS

LECTURE-06 **PHYSICS**

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Topics To Be Covered

- 1** Revision
- 2** Surface Tension ✓
- 3** Surface Energy ✓
- 4** Angle of Contact
- 5** Excess of Pressure inside the curve

RECAP FROM 4TH LECTURE

Pressure due to a fluid column; Pascal's law and its applications (hydraulic lift and hydraulic brakes), effect of gravity on fluid pressure. Viscosity, Stokes' law, terminal velocity, streamline and turbulent flow, critical velocity, Bernoulli's theorem and its simple applications. Surface energy and surface tension, angle of contact, excess of pressure across a curved surface, application of surface tension ideas of drops, bubbles and capillary rise.

RECAP FROM 1ST LECTURE

Fluid

liquid

gas

Density ρ

$$\rho = \frac{m}{V}$$

Pressure

$$P = \frac{F}{A}$$

Atmospheric pressure P_0

Variation of pressure with depth ✓



$$P_B = P_A + \rho gh$$

Pascal Law

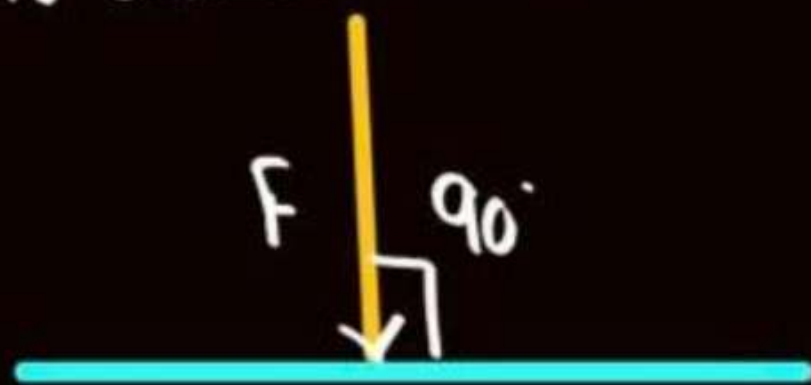
Recap-2

① Application pascal law

- Hydraulic Brake ✓
- Hydraulic lift

② Thrust

- Any perpendicular force is called thrust

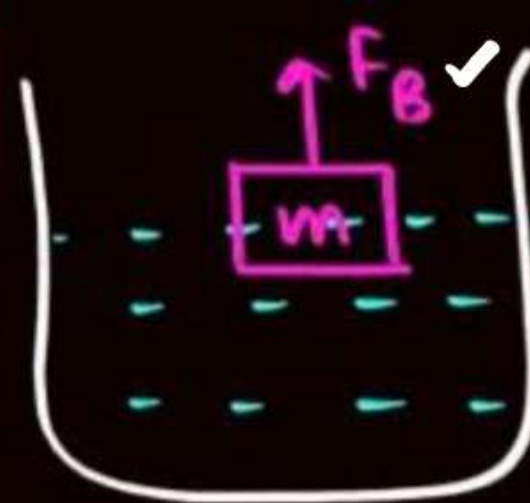


Buoyant force



Archimedes principle

$$F_B = \rho V g$$



Flow

Streamline flow

$$Re < 2000$$

Equation of Continuity

$$a_1 v_1 = a_2 v_2$$

turbulent flow

$$Re \gg \gg 2000$$

Recep - ③

Bernoullies Theorm

- law of conservation of Energy

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$$

Application of B-Theorm

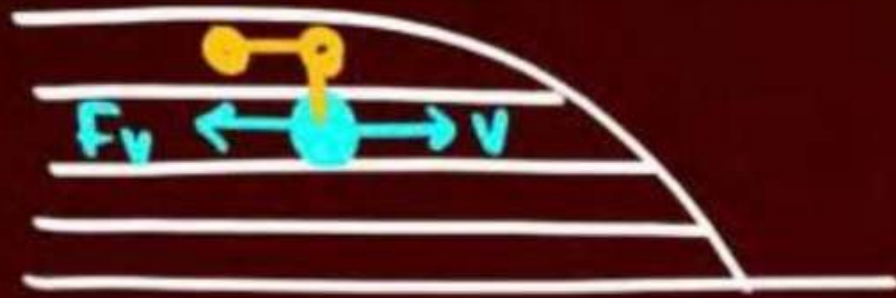
① Spray / Atomizer

② Magnus effect

③ lift of aeroplane

RECAP FROM 4TH LECTURE

• Viscosity



$$F_v = -\eta A \frac{dv}{dx}$$

$\eta \Rightarrow$ coefficient of viscosity

η = Pascal sec (SI)

η = poise (c.g.s)

Dimensional formula of η

$$\checkmark \eta = m l^{-1} t^{-1}$$

Stoke law

$$F_v = 6\pi\eta r v \checkmark$$

$$a=1$$

$$b=1$$

$$c=1$$

RECAP FROM 5TH LECTURE



- Terminal velocity



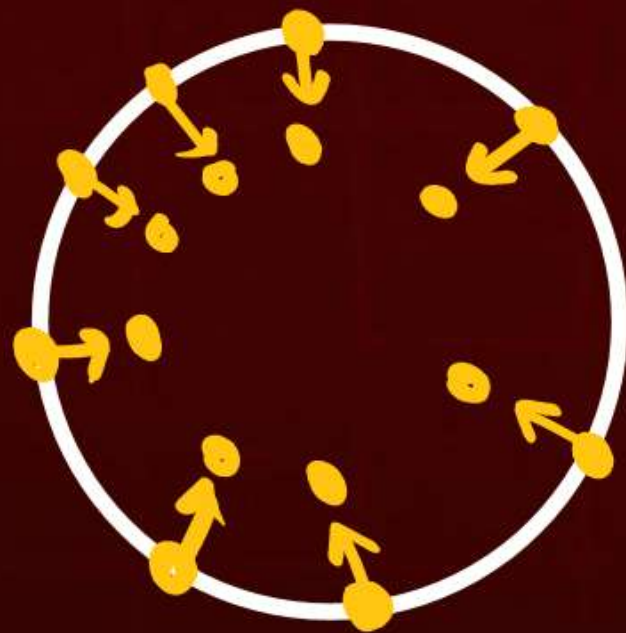
$$v_T = \frac{2}{9} \frac{r^2 g (\rho - \rho_f)}{\eta}$$

Some Important Phenomenon due to Surface Tension

①

(i) Shape of water droplet ✓

Spherical due to Surface tension

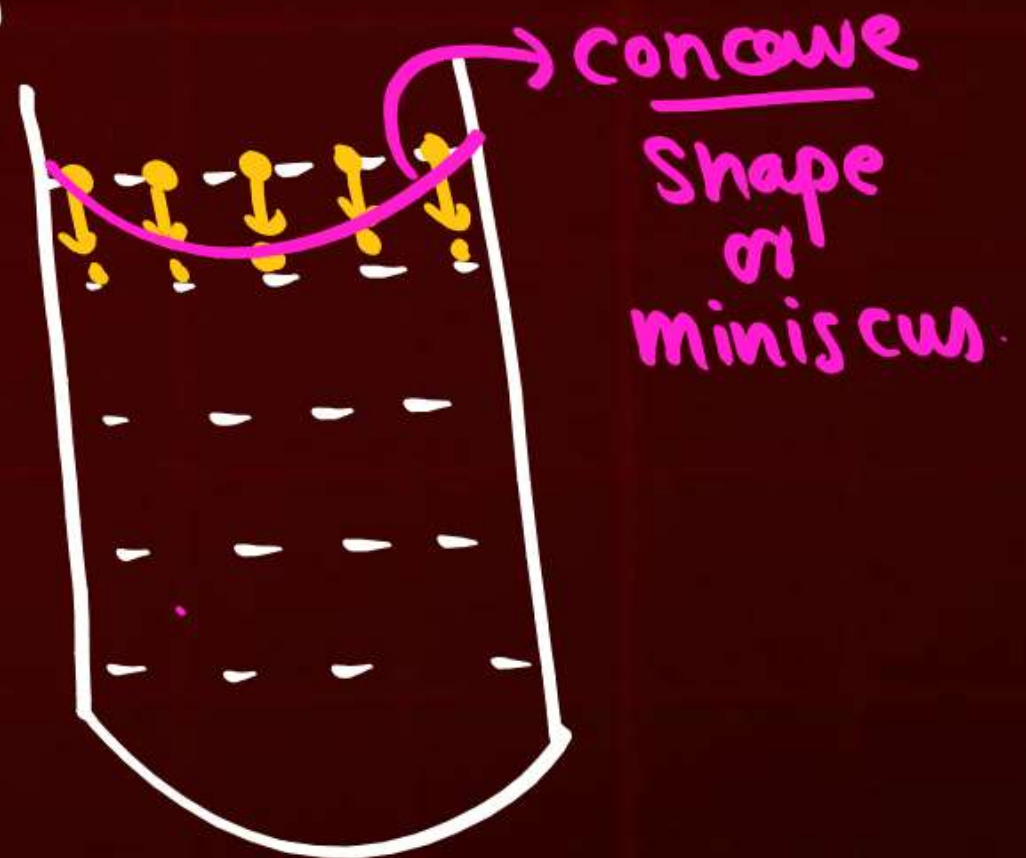


(ii) Meniscus of water surface

mca

concave

- The free surface of water appears in concave shape due to Surface tension





②

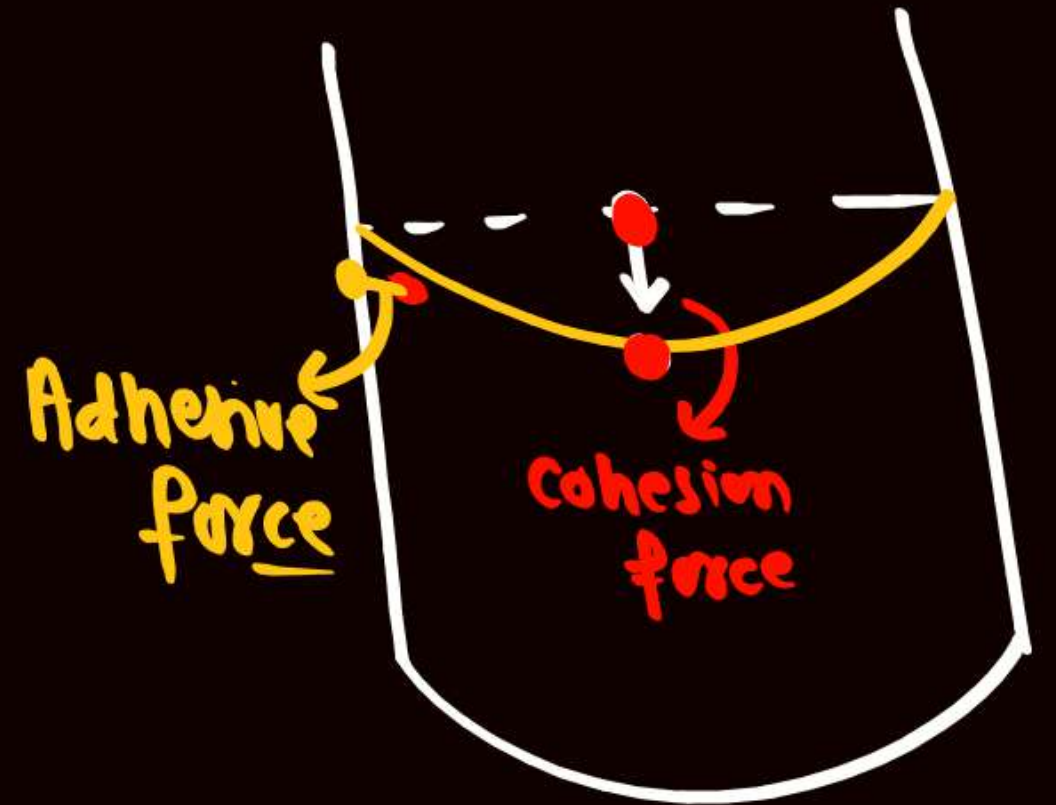
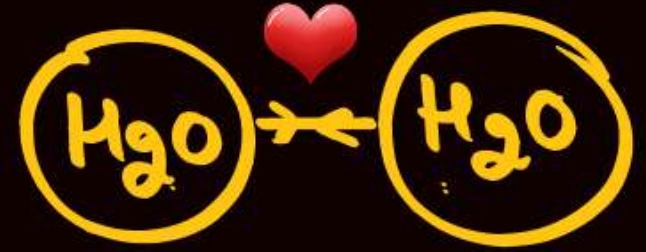
Note

Force of cohesion / cohesive Force

⇒ The force of attraction b/w same molecule

Force of Adhesion / Adhesive force

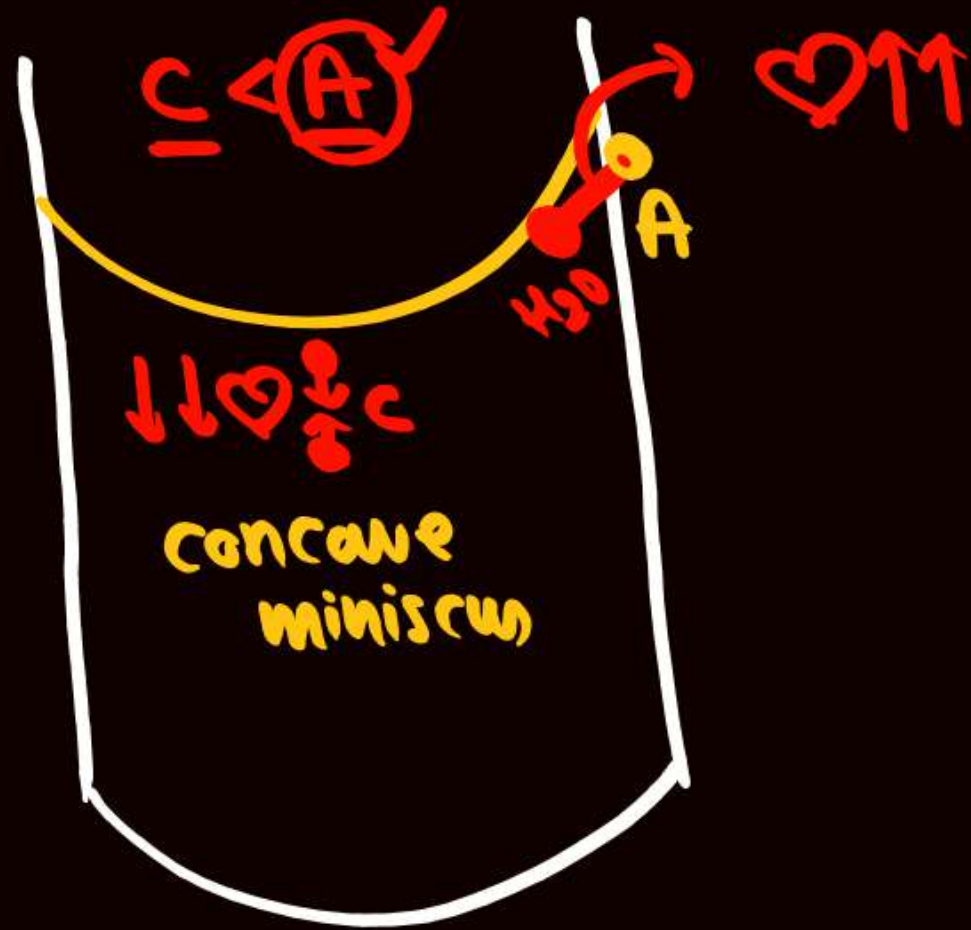
⇒ The force of attraction but
b/w two unlike molecule



Case - study

③

water



Hg



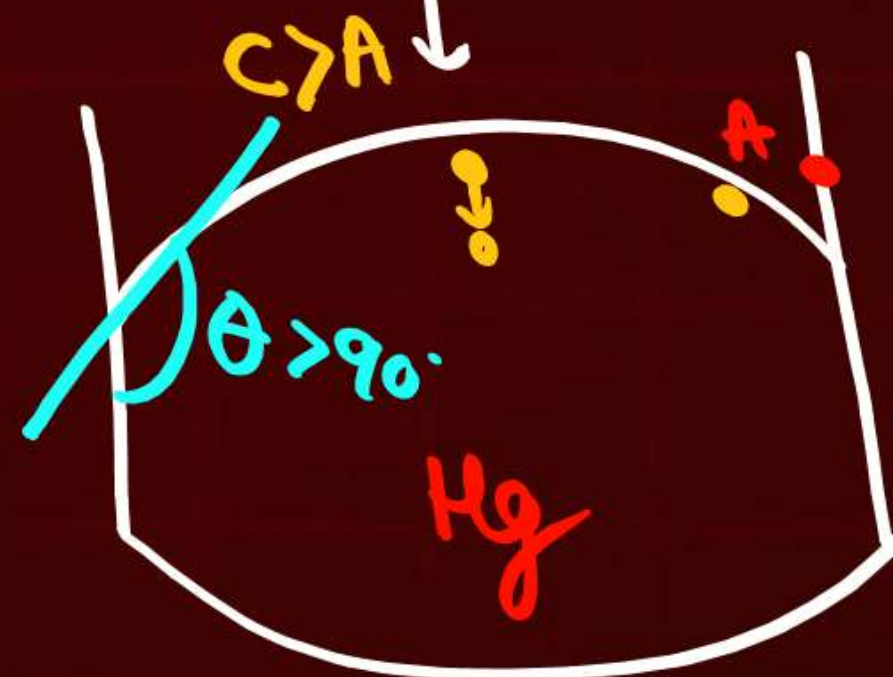
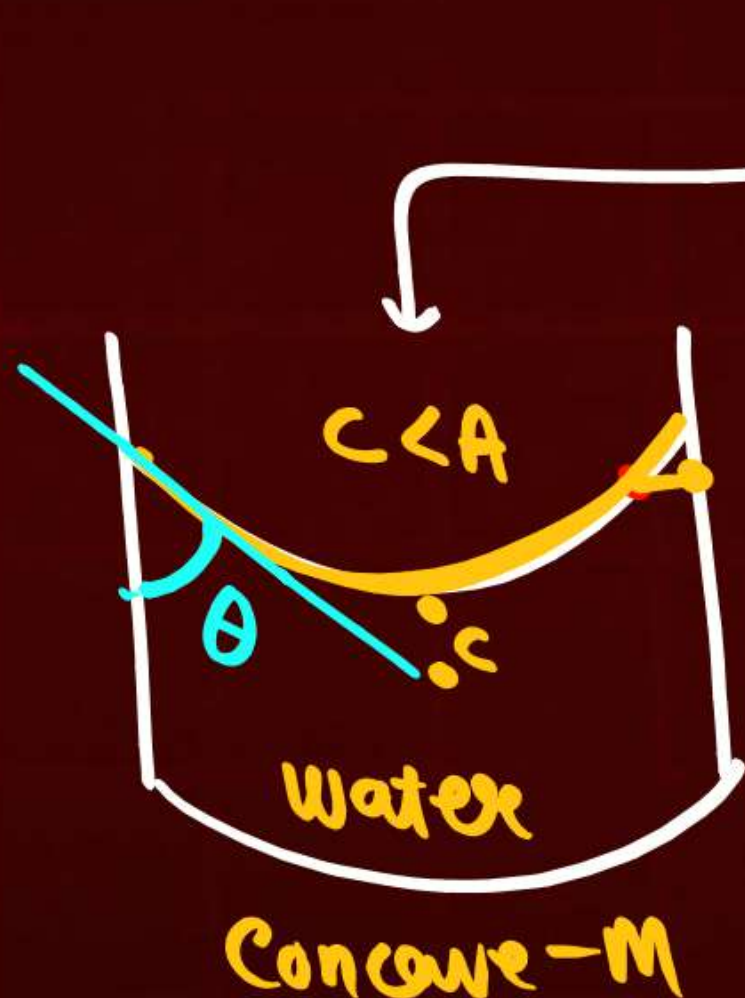
Meniscus

4



The Curved liquid Surface is called Meniscus.

Neet
2022



Note :- The liquid with Concave-M wet the substance

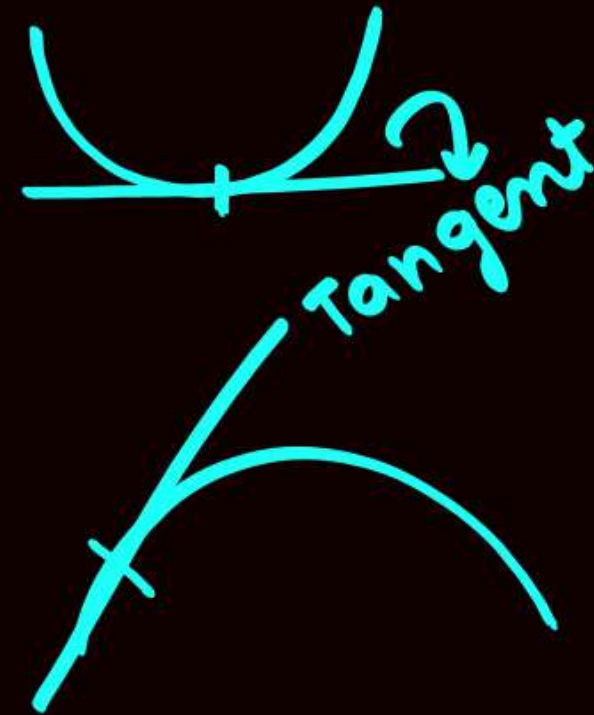
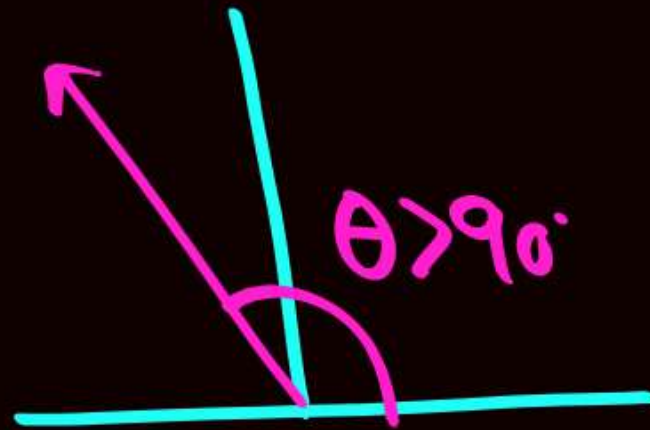
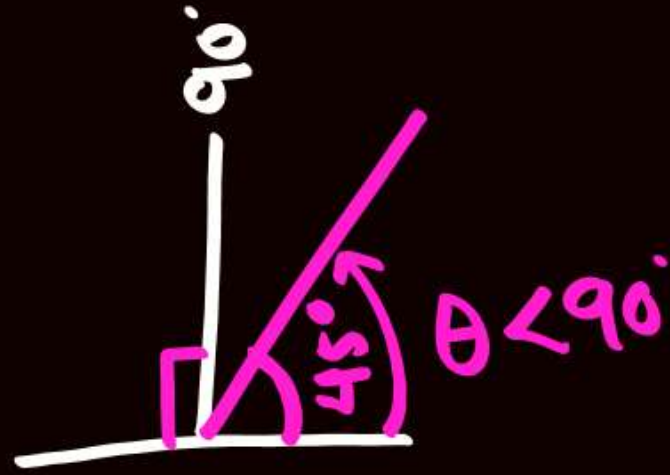


Acute Angle

angle less
than 90°

obtuse Angle

angle larger
than 90°

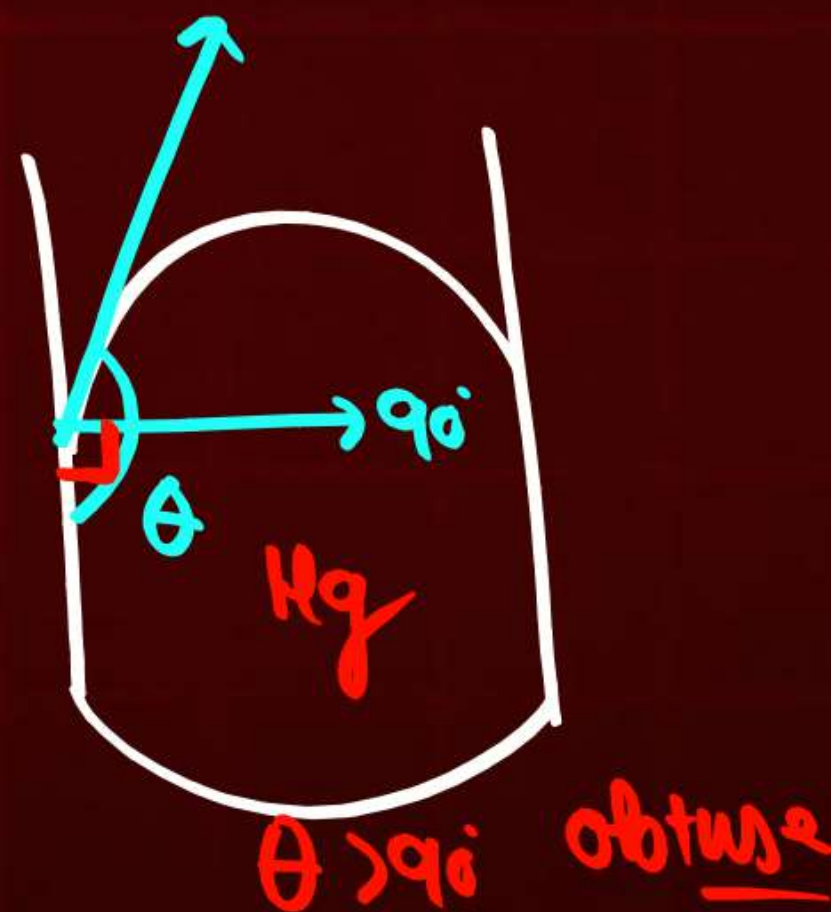
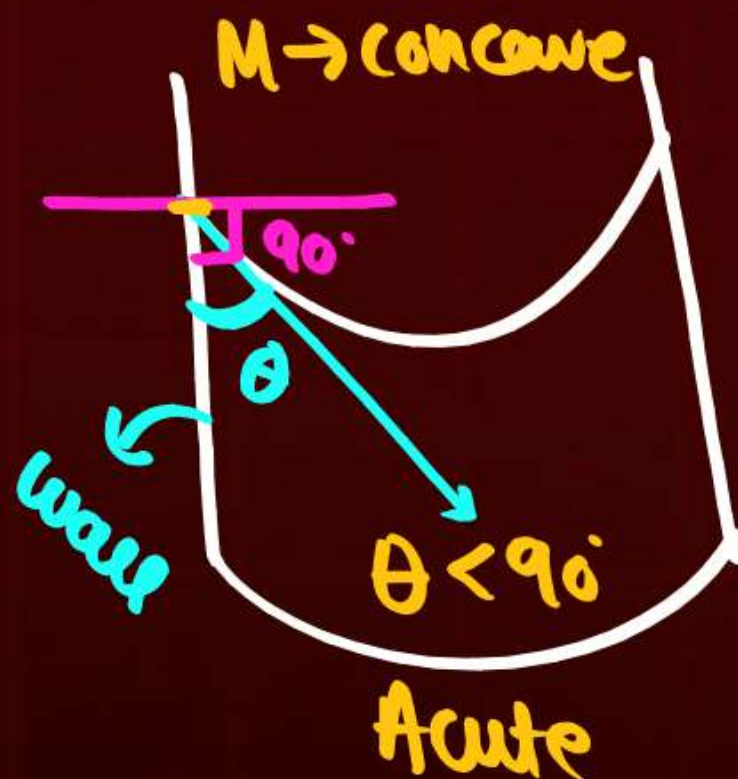


Angle of Contact

- It is another technique to find out the shape/miniscus of liquid surface

Defination

The Angle b/w tangent of liquid surface and wall inside the liquid



Excess Pressure

ΔP $\rightarrow$ delta
change

6



- The Excess pressure is the pressure difference b/w System (liquid) and surrounding



$$\Delta P = P_0 - P_A$$



2023

Excess pressure of water drop.

⑦

It is the pressure difference b/w surrounding and in drop



$$\text{Excess pressure} = P_B - P_A = \Delta P$$

$$\Delta P \propto T \quad \text{--- ①}$$

$$\Delta P \propto \frac{1}{R} \quad \text{--- ②}$$



$$\Delta P \propto \frac{T}{R}$$

$$\Delta P = K \frac{T}{R}$$

$$K = 2$$

$$\Delta P = \frac{2T}{R}$$

Numerical

Q. A drop of 2m Radius having the surface tension = 4N/m
Calculate the Excess pressure

⑧

$$m=2$$
$$a=5$$

$$F=ma$$
$$=2 \times 5$$
$$=10N$$

(a) 2P

(b) ✓ 4P

(c) 8P

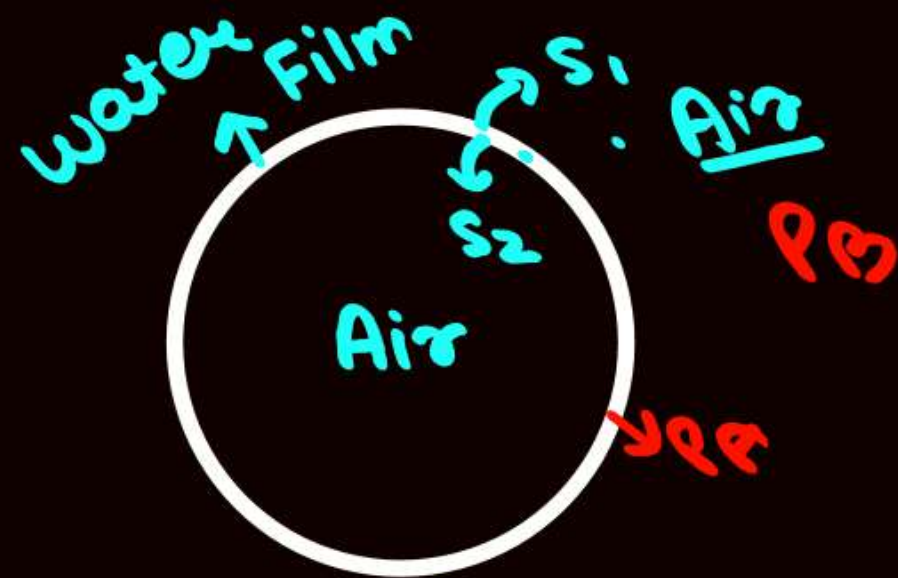
(d) 16P

$$\Delta P = \frac{2T}{R}$$
$$= \frac{2 \times 4}{2}$$
$$= 4$$

(II) Excess pressure in case of bubble

⑨

It is a pressure difference b/w surrounding and bubble

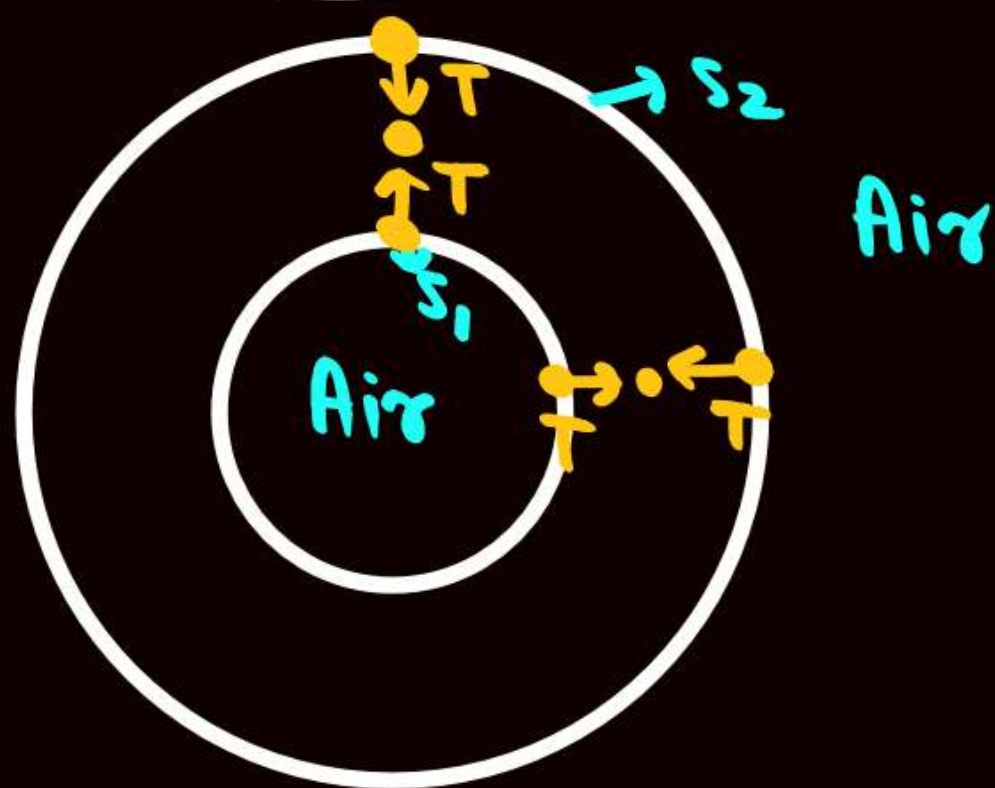


$$\Delta P = P_B - P_A = \text{Excess pressure}$$

$$\Delta P \propto 2T \quad \text{--- ①}$$

$$\Delta P \propto \frac{1}{R} \quad \text{--- ②}$$

$$\Delta P = \frac{4T}{R}$$



$$\Delta P \propto \frac{2T}{R}$$

$$\Delta P = \frac{K 2T}{R}$$

$$\Delta P = 2 \left(\frac{2T}{R} \right) \Delta P_0$$

Numerical

Q. A bubble of 2m Radius having the surface tension = 4N/m
Calculate the excess pressure

(a) 2P

(b) 4P

(c) 8P

(d) 16P

$$\begin{aligned}\Delta P &= \frac{4T}{R} \\ &= \frac{4 \times 4}{2} \\ &= 8 \text{ pascal}\end{aligned}$$

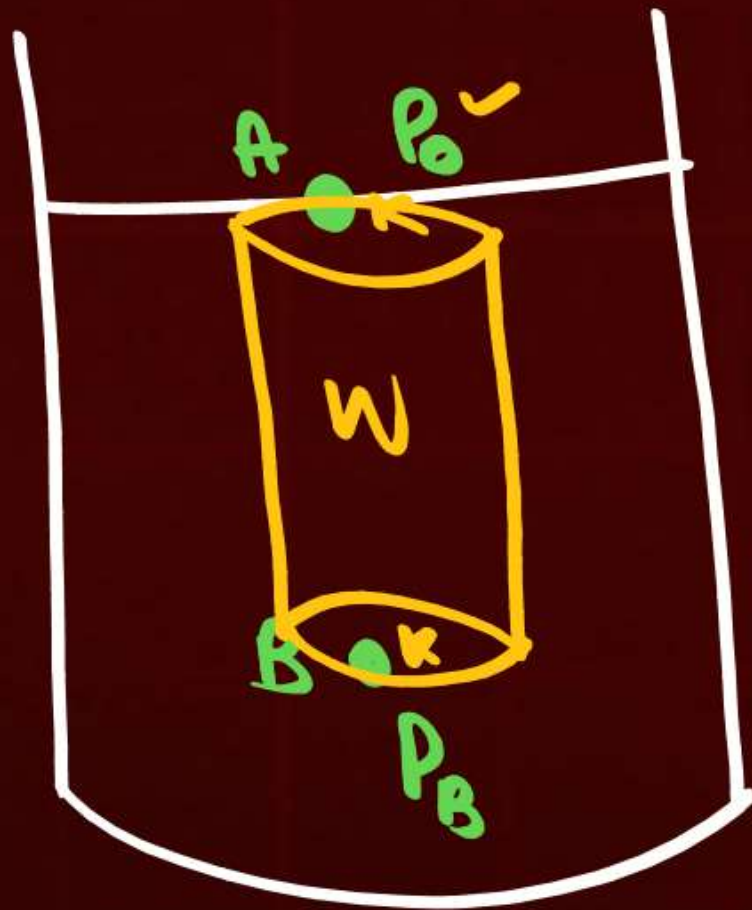
(10)

$$\begin{aligned}m &= 2 \\ a &= 5\end{aligned}$$

$$\begin{aligned}F &= ma \\ &= 2 \times 5 \\ &= 10 \text{ N}\end{aligned}$$

✓ Effect of (g) on pressure

✓ Variation of Pressure with depth



$$P_B > P_A$$

$$P_B = P_0 + \rho g h$$

$$3 = 1 + 2$$

$$4 = 1 + 3$$

Pressure increases when g increases



Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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